Mixed Quant Strategies in Crypto Trading

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Introduction

Overview

In this project, we will examine and evaluate several different quantitative trading strategies for cryptocurrencies.

All strategies assume fees of 20bps per trade.

Strategies

1. A technical indicator based strategy
2. An ML model trained to predict returns
3. An ML model trained to identify bullish/ bearish regimes.

Goal

The goal is to build a mixed, and more robust trading strategy by using different approaches.

Benchmarks like Buy & Hold, and comparing to BTCUSDT (used as beta) are used to assess performance.

Data and asset selection

DATA FREQUENCY	HISTORICAL DATE RANGE	VALIDATION DATE RANGE	OOS DATE RANGE
Daily	2021-01-01 to 2023-12-31	2024-01-01 to 2024-12-31	2025-01-01 to 2026-02-28

MODEL TYPE	DATA SOURCE(S)	ASSET UNIVERSE
Technical Indicator Model	Binance Python API	Top 100 coins based on market share (*coinmarketcap.com*)
ML Return Prediction Model	Binance Python API	Top 100 coins based on market share, filtered to subset of coins that have enough data in ***historical date range*** for training
ML Regime Identification Model	Binance Python API	Top 100 coins based on market share, filtered to subset of coins that have enough data in ***historical date range*** for training

Strategies

Technical Indicator Strategy

Description → This model aims to capture both ***LONG AND SHORT*** trading signals from traditional technical indicators.

Technical indicators → Moving averages, momentum, relative strength index, drawdowns, and others engineered from these.

Volatility filter → Volatility is used to derive a separate indicator that stops trading during certain conditions.

Strategies

ML Return Prediction Strategy

Description → This strategy takes a machine learning approach to finding trading opportunities by predicting next day returns. This is a ***LONG POSITION ONLY*** strategy.

Algorithm → Uses momentum features and performs logistic regression on next day returns. The model classifies the target into groups that are either above or below a chosen threshold.

Optimization → It is optimized for precision, since in trading, false positives lead to losses, whereas false negatives cost nothing.

Strategies

ML Bullish/Bearish Regime Identification Strategy

Description → This is another machine learning approach that attempts to identify periods of ***LONG AND SHORT*** opportunities.

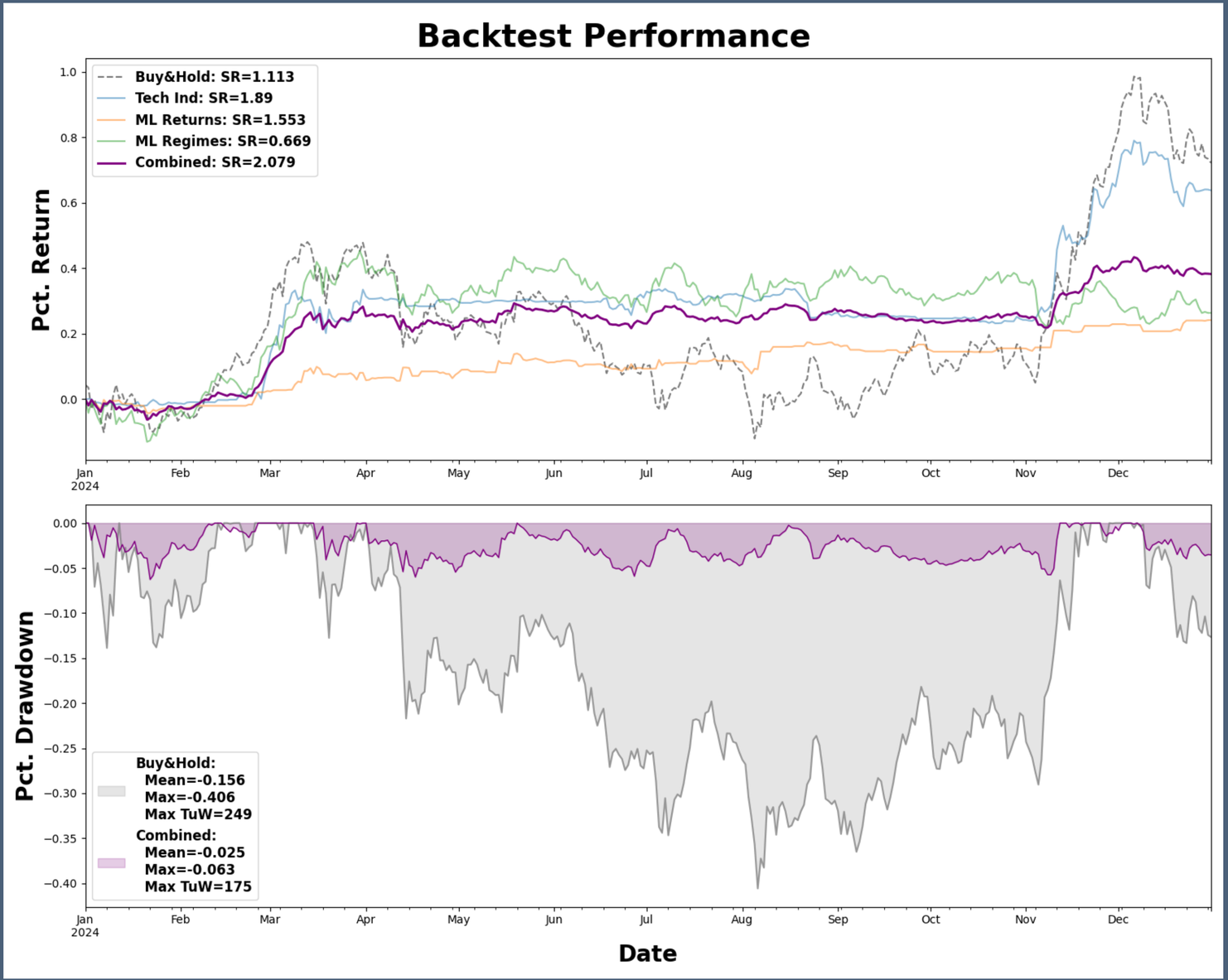
Algorithm → Like the returns prediction model, uses momentum features as input. This is an unsupervised model that uses principal component analysis to reduce dimensionality, and clusters data points into optimal long-short-neutral regimes.

Grid Search → A grid search is used to determine the optimal number of principal components, and number of clusters. Sharpe ratio is used as evaluation metric in grid search.

Backtest performance

- All but regime model outperforms Buy & Hold benchmark in Sharpe ratio
- Without prior performance to use for weights, these models were combined equally, obtaining a **Sharpe ratio of 2.079.**
- Drawdown for combined strategy outperforms Buy & Hold benchmark in mean, max, and max time-under-water

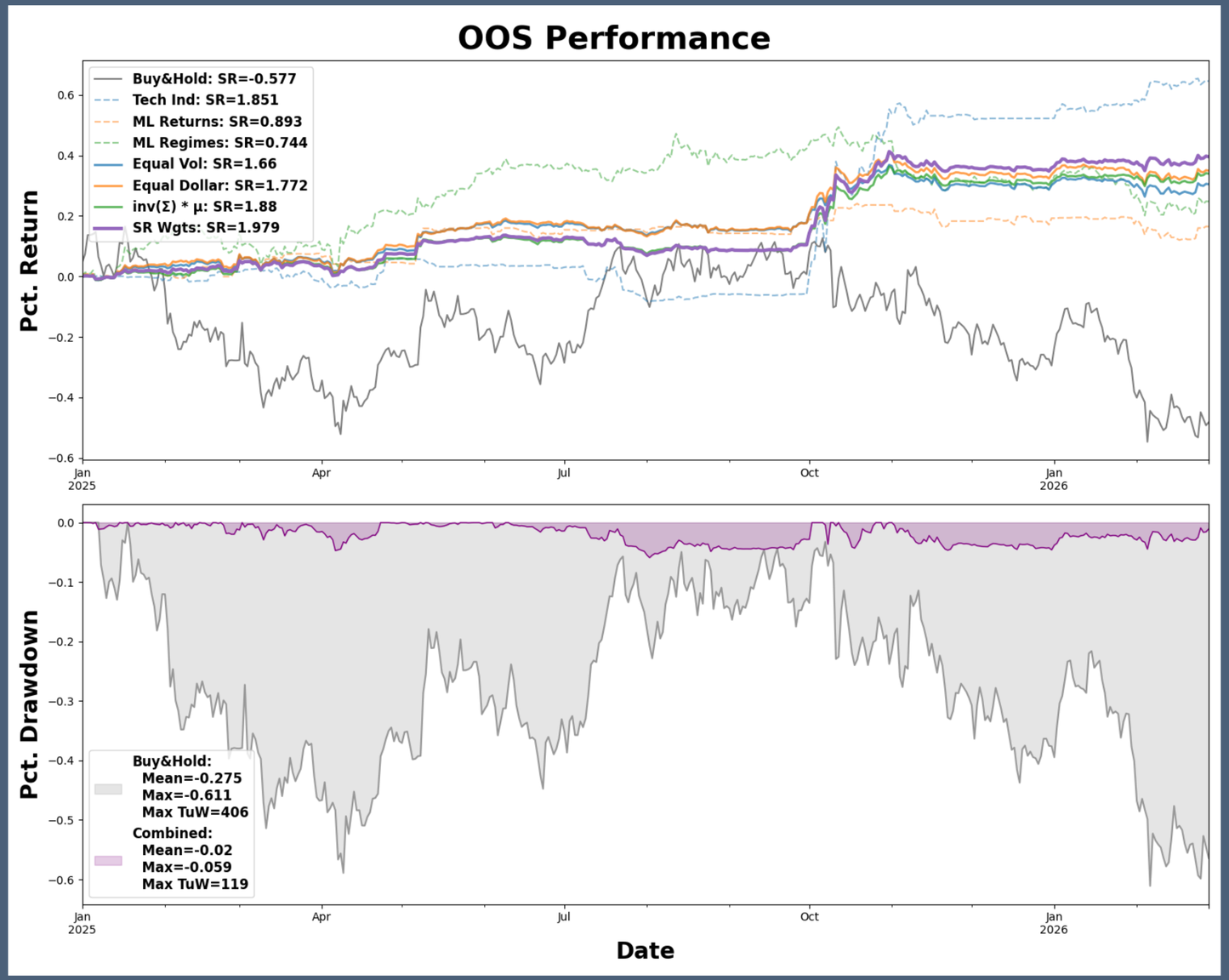
	Tech. Ind.	ML Returns	ML Regimes	Combined
Tot. Returns	0.638	0.241	0.263	0.382
Volatility	0.0176	0.0081	0.0206	0.0096
Alpha coeff.	0.0016	0.0003	0.0002	0.0010
Alpha t-stat	1.171	0.737	0.187	1.926



Out-of-sample performance

- Tested various weight combinations
- Combining strats based on their Sharpe ratios during validation period is best performing in OOS period: **Sharpe Ratio 1.979**
- Drawdown for Sharpe ratio weighted strategy outperforms that of the Buy & Hold benchmark

	Equal Volatility Weights	Equal Dollar Weights	inv(Sigma) * mu	Sharpe Ratio Weights
Tot. Returns	0.305	0.350	0.340	0.396
Volatility	0.008	0.009	0.008	0.009
Alpha coeff.	0.0007	0.0008	0.0008	0.0009
Alpha t-stat	1.772	1.895	2.019	2.126
Turnover	0.03	0.07	0.04	0.07
Information Ratio	0.92	0.97	0.97	1.03



Conclusions

- The three individual strategies utilize much of the same data, but their diversified approaches in capturing trading signals lead to a more robust combined model with higher Sharpe ratio.

- **Individual Strats:**

  i. Technical indicator strategy stays flat for long periods, but can capture quick periods of high returns.

  ii. ML Return Prediction model sees lower gains, but lower volatility compared to other models, due to its optimization to avoid false positives.

  iii. ML Regime Identification model doesn't perform as well as others, but is consistent across validation and OOS periods. Still useful to combine with others to increase net Sharpe ratio.

- **Combined Strat:**

  i. The combined strat outperforms the Buy & Hold benchmark in Sharpe ratio, and drawdowns.

  ii. Alpha coeff 0.0009 and alpha t-stat 2.126

  iii. Turnover for the combined strat is ~7% daily on average.